

Next Generation Networks Architecture and Layered End-to-End QoS Control*

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Abstract. Next-generation network (NGN) is a new concept and becoming more and more important for future telecommunication networks. This paper illustrates five function layers of NGN architecture and discusses some end-to-end QoS (quality of service) issues for NGN (called NGNQoS). The five function layers are: (1) Application Layer that supports SIP protocol; (2) Network Control Layer that aims at overcoming the bottleneck problems at edge nodes or servers for end-to-end admission control; (3) Adaptation Layer that supports different network configurations and network mobility; (4) Network Transmission Layer that provides end-to-end QoS control for real-time communications through integrating Differentiated Service (DiffServ) and Multi-Protocol Label Switching (MPLS) and (5) Management Layer that provides Web-based GUI browser for data presentation, monitoring, modification and decision making in NGN.

1 Introduction

Next-generation network is a new concept commonly used by network designers to depict their vision of future telecommunication networks. Various views on NGN have been expressed by network operators, manufacturers and service providers. NGN seamlessly blends the end-to-end QoS into the public switched telephone network (PSTN) and the public switched data network (PSDN), creating a single multi-service network, rather than a large, centralized and proprietary infrastructure. Next-generation network architecture pushes central functionality to the edge of the network. The result is a distributed network infrastructure that leverages new, open technologies to reduce the cost of market entry dramatically, increase operational flexibility, and accommodate both circuit-switched voice and packet-switched data services. The integrated services will bring communication market billions of incomes, however, the R&D for NGN still lack behind the actual demands of the society [1]. On the other hand, the architecture of the Internet and IP-based networks is rapidly evolving towards one where service-enablement, reliability and scalability become paramount.

Dynamic IP routing supported by routing protocols such as OSPF, IS-IS and BGP provides the basic internetworking function while confronting the dual challenges of larger scale and faster convergence. Many providers are looking to a converged

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packet switching network (PSN) based on IP/MPLS. The transport layer protocols including TCP and SCTP continue to be an area of active research as developers seek optimal application throughput and resilience. IP QoS defined by IntServ and DiffServ continues to evolve and interesting efforts are underway to enhance QoS signaling for both wired and wireless networks.

The challenges and opportunities associated with a fundamental transformation of current networks toward a multi-service ubiquitous infrastructure with a unified control and management architecture have been discussed in [2], which presented the outline of the fundamental reasons why neither the control infrastructure of the PSTN nor that of the present-day Internet is adequate to support the myriad of new services in NGN. Although NGN will inherit heavily from both the Internet and the PSTN, its control and management architecture is likely to be radically different from both, and will be anchored on a clean separation between a QoS-enabled transport/network domain and an object-oriented service/application domain, with a distributed processing environment that glues things together and universally addresses issues of distribution, redundancy, and concurrency control for all applications.

This paper presents NGN architecture and discusses the layered end-to-end QoS control for NGN. In Section 2, a survey for NGN is given and the five function layers of NGN are illustrated in Section 3. Some end-to-end QoS issues in NGN are described in Section 4 and we conclude in the final section.

2 Survey of NGN: Research and Development

Telcordia Technologies in NJ, USA proposed next generation networks that support a variety of communication services (data, video, and voice) seamlessly [4]. Customers will demand that these networks be highly reliable as there will be more and more traffic and services. Because of the historically exceptional reliability of wireline voice telephony, the reliability of voice services supported by NGN necessitates special attention in order to achieve the customer satisfaction of the service.

In South Koera, KT is considering the installation of NGN backbone network. QoS discussions on whether the IP router satisfies the forthcoming NGN customers who use basic application of NGN still remain. QoS values as packet delay, packet loss and jitter are measured and analyzed at the KT-NGN test bed, and are compared with the ITU-T QoS recommendation values [5].

Some German companies discuss QoS from a somewhat unconventional point of view and argue that high availability is a key ingredient in QoS perceived by the user. High availability with extremely short interruptions in case of failure is needed for acceptable QoS in real-time dialog services such as telephony or video conferencing and an even distribution of the traffic load over the network is essential to ensure the efficient network utilization given that some kind of admission control for QoS traffic has to be in place for overload avoidance [9].

Alcatel (France) proposes the NGN multimedia network structure and its business model with four players involved in charging: access provider, connection provider, telecommunication service provider, and value-added service provider. Often charging components must be correlated to create a clear postpaid bill and ensure

correct treatment of prepaid accounts, as well as settlement between the providers involved. If charging is to remain a prime competitive tool in next-generation networks, it must be functionally intelligent and flexible, and able to optimize network operator and service provider revenues while providing a fair policy toward the end users [10].

In UK, next generation IP-based networks that offering QoS guarantees by deploying technologies such as DiffServ and MPLS for traffic engineering and network-wide resource management have been proposed. An ongoing work towards inter-domain QoS provisioning is presented [16].

The basic issue of NGN trials on Russian public networks is interoperability testing of foreign equipments that are adapted to Russian network, domestic NGN system SAPFIR. Results of these NGN trials will be used for the development of the “NGN Evolution Concept” for Russian public networks [17].

3 Overall NGN Architecture

NGNQoS can be described from five function layers: (1) Application layer that contains the typical middleware for authorization, accounting, directory, search and navigation for millions of users; (2) Network control layer aims at overcoming the bottleneck problems at edge nodes or servers and it is composed of a series of control

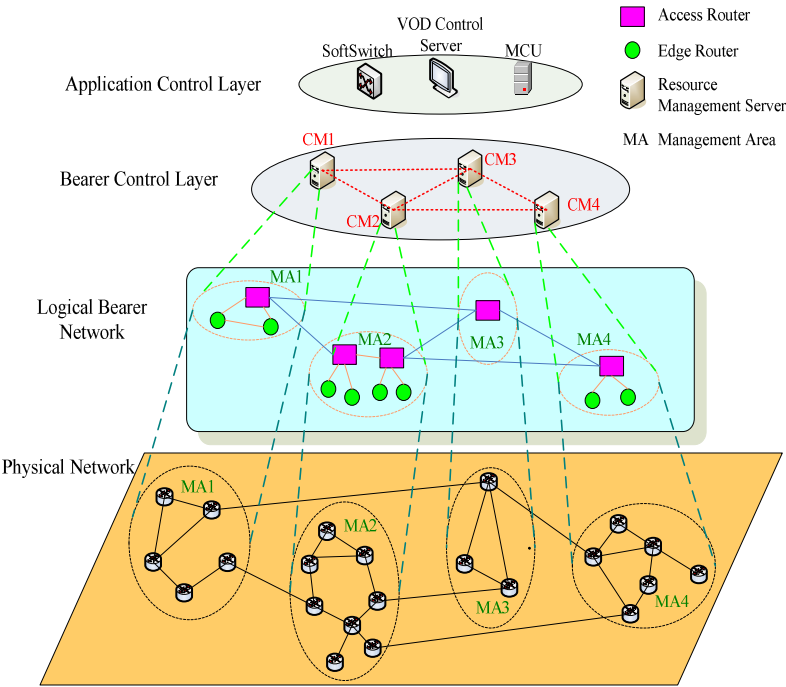


Fig. 1. NGN Network Architectures

agents for admission control, call setup and end-to-end QoS control through available bandwidth detection, local information control, class priority and intelligent scheduling. Multicast and anycast group managements will be implemented to leverage the load for admission control or service/message distributions; (3) Adaptation layer that supports different network configurations and network mobility. This layer can provide soft switching between different networks on different levels such as IPv4, IPv6, ATM, Ethernet, WLAN, WMAN and 3G networks. It supports both packet and circuit switching and provides interconnection between the two switching networks; (4) Network Transmission Layer that provides the effective end-to-end QoS control for real-time requests and flows through integration of parameterized QoS control and class priority control. This is particularly important to resolve the bottleneck problems such as multi-path routing that enables the multiple choices for the path and anycast routing that enables the selection from different (replicated) servers and (5) Management layer that provides Web-based GUI browser and wireless connection information such as the data access using XML and Web-based visualization for data presentation, monitoring, modification and decision making in NGN. The IP telecommunication network architecture and software layer architecture are shown in Fig. 1 (see <http://www.huawei.com>) in which Bearer Control Layer and Logical Bearer Network perform network control together.

4 Layered End-to-End QoS

This section describes the details of each layer and their functions for layered end-to-end QoS control. Note that we do not intend to give all the functions for NGN layers but give some important QoS issues and introduce our designs and algorithms.

4.1 Application Layer: SIP

Application layer contains typical middleware for authorization, accounting, directory, browser, search and navigation for millions of users. Web services have been discussed extensively; however, there are not many discussions about end-to-end service on the NGN architecture, especially with mobility and multimedia transmission supported. In this subsection, we only focus on a prototype that can provide wired and wireless QoS service using Session Initiation Protocol (SIP) [11].

SIP is an application layer signaling protocol which is used for managing multimedia sessions among different parties. The principle role of SIP is to set up sessions or associations between two or more end users. Initiated Sessions can be used to exchange various types of media data using appropriate protocols such as RTP, RSTP. Currently, SIP is able to set up a call carrying the information of a more detailed multimedia session using protocols such as the Session Description Protocol (SDP) [21]. By using adaptive protocol, the selection mechanism is achieved through applying the most suitable protocol for end user devices during communication without any interruption and disconnection [22]. The SIP implementations in application layer may integrate both wired and wireless networks based on NGN architecture.

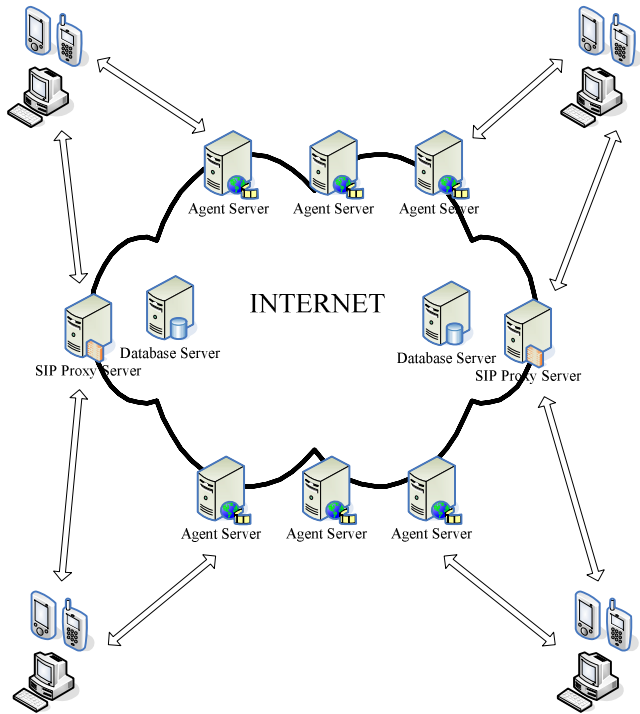


Fig. 2. SIP based end-to-end multimedia transmission system

Little work is done to enable end-to-end QoS multimedia transmission over hybrid of wired and wireless networks with SIP. Based on SIP, we have implemented an end-to-end multimedia transmission system, called AnyServer, for real time and non-real time video/audio communications, as shown in Fig. 2. To achieve SIP based end-to-end multimedia transmission, SIP is not only used for call setup signaling, but also carries information for session establishment in adaptive protocol selection mechanism. SIP carries an SDP packet describing an audio or video session, indicating supported communication protocols and end terminals' capabilities. To select the most suitable protocol for adapting different situations intelligently during a communication, data buffering service is also provided. In this way, end users can communicate with the others at their best acceptable QoS level. Currently, we are integrating AnyServer with NGN to provide multi-point end-to-end QoS applications such as video conferencing. QoS requirements of applications and session IDs are used for user identification of multi-parties communication in video-conferencing. Four major functional components of the current system are User Agent in client device, SIP Proxy Server, Database Server and Agent Server to form the heterogeneous wireless and Internet services [15, 22].

4.2 Network Control Layer

Network control layer is composed of a series of control agents for distributed admission control (DAC), call setup and end-to-end QoS control through available bandwidth detection, local information control, class priority and intelligent scheduling. We discuss this layer based on the following functions:

(1) Traffic classification for incoming requests: This function is performed by scheduler agents that examine the legal incoming requests and make classifications. The classified traffic will be processed through admission control agents (or the admission nodes). To avoid any unnecessary delay, the scheduler and the admission control agents normally reside on the same site. By this approach, the Internet and telecommunication tasks can be classified and treated properly as detailed in the end-to-end QoS design.

(2) Admission control: We have designed admission control algorithms which perform bandwidth detection and connection control. Bandwidth detection enables the approximate network resources to be detected in case that the networks are managed by different administrators or involved in heterogeneous networks. Based on the available bandwidth detection and class priority for incoming requests, our distributed admission control algorithms can enhance the scalability and admission probability. A cooperative distributed approach can be implemented at some board NGN admission nodes and we give a brief discussion of admission control algorithm for anycast flows [6] below.

Anycast flow is a flow which may connect to any destination in a target group. We consider anycast flow as a general flow concept because the anycast flow may be a unicast flow if the group only has one destination or multicast flows when the flow must be sent to every destination in the group. We first consider the destination selection issue. Destination selection determines which destination the anycast flow should be sent to. A good selection will bring a better chance for the flow to be admitted. We propose several weight assignment algorithms based on available information such as route distance and available bandwidth. Different status information surely impacts differently on the network performance in terms of admission probability, overhead, and compatibility as illustrated below:

(1) Weighted destination selection based on the static route distance information [6, 20]: The admission control routers/servers may apply even weight assignments for the destination selection if none of the information is available. The length of the route may be easily obtained via current routing protocols [18, 19]. The differences of route distances reflect the different resource consumption by the anycast flow. Intuitively, the flow to destinations with shorter distances will consume less bandwidth and fewer resources. Hence a smart destination selection algorithm should prefer destinations with short route distances.

(2) Weight assignment based on local admission history: The local admission history may be defined as a log that records the successfulness of selecting individual destinations in admission control. Let H_i record the number of the continual failures in the most recent admission history. For example, $H_i = 3$ implies that for the last three times when destination i was selected in admission control process, there was insufficient bandwidth and resource reservation. We proposed a destination selection algorithm to combine both route distance and local admission history information and the admission probability is expected to be higher than that of some static algorithm.

(3) Weighted Assignment based on available bandwidth: Local admission history may not accurately reflect the network dynamic status. We may also use available bandwidth detection for admission control. Resource Reservation can be made by some standard protocols such as RSVP [12] or by checking the availability of link bandwidth along the route based on the approach illustrated before. We have extended our anycast admission control protocol [3, 6] to include the available bandwidth information.

4.3 Adaptation Layer

This layer provides soft-switching between different networks on different levels such as IPv4, IPv6, ATM, Ethernet, WLAN, WMAN or 3G networks which support both packet and circuit switching. The layer can be divided into the following major functions:

- (1) Soft switching between IPv4 and IPv6 using tunneling techniques carried out by edge routers of the subnet between the networks.
- (2) ATM convergence sub-layer merges the ATM cells to IP packets (which may be used by WLAN and WMAN networks).
- (3) Soft switching between ITU H.323/H.324 protocols to handle the circuit/packet switching.

We have efficiently implemented the 3G-324M protocol stack for 3G wireless communications [7]. Fast transformation between circuit switching networks to packet switching networks is under development. We are currently designing some new algorithms for the connections of heterogeneous wireless networks such as WLAN, WMAN and 3G networks.

4.4 Network Transmission Layer

In this layer, we focus on the discussions of Differentiated Service and Multi-Protocol Label Switching and the related end-to-end QoS issues.

Differentiated service. Differentiated Service architecture achieves scalability by aggregating traffic classification state. Packets are classified and marked to receive a particular per-hop forwarding behavior on nodes along their path. Sophisticated classification, marking, policing, and shaping operations only need to be implemented at network boundaries or hosts. The major advantage of DiffServ is that the Internet flows can be differentiated from the telecommunication flows by the board routers that may deal them with different QoS requirements. This is particularly useful for NGNQoS. We designed some special devices called network mapping (NM) that maps user's QoS requirement into service level agreements contract between customer and Internet service provider (ISP). The admission control can also be integrated with DiffServ architecture for the end-to-end QoS solutions. Fig. 3 shows the block diagram of a classifier and traffic conditioner. Note that a traffic conditioner may not necessarily contain all the four elements. For example, in the case where no traffic profile is in effect, packets may only pass through a classifier and a marker.

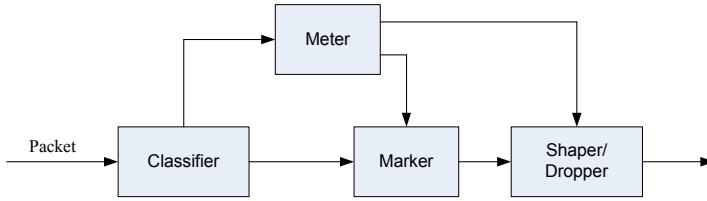


Fig. 3. Logical View of a Packet Classifier and Traffic Conditioner

To achieve the scalability and QoS for the DiffServ flows, we have designed a generalized regulator to provide an adaptive traffic control mechanism for very high rate real-time aggregated flows, especially, for those traffic that have been marked as red. Normally, three classes of traffic flows (green, yellow and red) in DiffServ network are defined in [13, 14] and we are interested in the deterministic delay bound for the real-time flows which may be marked as red/yellow but have stringent delay requirements. The generalized regulator, based on the extended network calculus, is developed for the purpose of effective control of high rate flows with QoS requirements as detailed in [8].

Multi-Protocol Label Switching (MPLS). In MPLS, packets are encapsulated at ingress points. The local significant labels, which have short fixed-length, are used in the headers of encapsulated packets. The packets are forwarded via Label Switching Routers (LSRs) by label swapping. An explicit path for each connection is called Label Switched Path (LSP). A reservation protocol is required to establish a LSP through a network. MPLS networks provide QoS guaranteed services with a lower computational complexity and operational costs, compared with IP networks using ATM connectivity structure. The most important advantage of MPLS networks is that they can perform the traffic engineering for load balancing, which is able to improve the network performance in a long run. Traffic engineering (TE) is in general the process of specifying the manner in which traffic is treated within a given network. Users usually expect certain performance from the network, which in turn should attempt to satisfy these expectations. The expected performance depends on the type of traffic the network carries, and is specified in the service level agreement contract between customer and ISP. The network operator, on the other hand, should attempt to satisfy the user traffic requirements. Hence, the target is to accommodate as many traffic requests as possible by optimally using the available network resources.

4.5 Management Layer

This layer provides Web-based GUI browser and wireless connection information such as the data access using XML. Web-based visualization presentation is critical for the management of NGNQoS for data presentation, monitoring, modification and decision making. Network management is an indispensable building block in our proposed NGN architecture. Effective management of the NGN is becoming the key to the successful competition and continued growth. NGN management layer contains the management functions relating to QoS, security and network management. There are five levels in NGN Management Layer defined as (1) fault-management level, (2) configuration level, (3) accounting level, (4) performance level and (5) security level.

Based on a modular concept of element management and domain management, we have designed the NGN Management Layer that fully supports day-to-day operation, administration and maintenance tasks, network configuration and service provisioning (including mass deployment for China Mobile in FoShan). We also plan to integrate NGN Management Layer into cross-domain management systems and the business processes of the network operators.

5 Conclusions

We have discussed some important design issues for the next generation architecture and different layers for end-to-end QoS control. The issues of NGNQoS presented in this paper are not exhaustive; however, the most functions presented are drawn from our design and implementation experiences. We are currently focusing on the implementation of cross layer platform of packet switching and circuit switching for 3G wireless networks. The future work will tackle with the cross layer protocols that can harness the heterogeneous networks across Internet, telecommunication and wireless networks.

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